

Project Synopsis:

To briefly summarize my goals and plans. The main thing I created was a timelapse of showing a map of rainfall and car crashes in Montgomery for the entire year of 2025. My goal is to look for a correlation between rainfall and the number of crashes.

The Data:

In pursuit of my primary goal, I have two datasets. The one with which I will do the most work is a CSV (comma-separated values) file, which I requested from NOAA (National Oceanic and Atmospheric Administration). I have data for all of Maryland, but I have filtered only the information from stations in Montgomery County. The tool for requesting weather data from NOAA is linked [here](#)

My second dataset is from Data Montgomery and contains information on car crashes within Montgomery County. It is available through data Montgomery's portal [here](#)

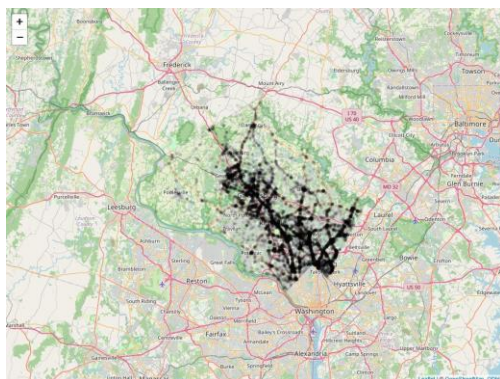
The Tools:

- R Studio (Interface)
- Leaflet (for maps)
- Gstat (for interpolation)
- Lubridate (for manipulation of DateVariables)
- Mapshot (for capturing images of maps)
- FFmpeg (for compiling images into a video)

Data Ingestion Summary:

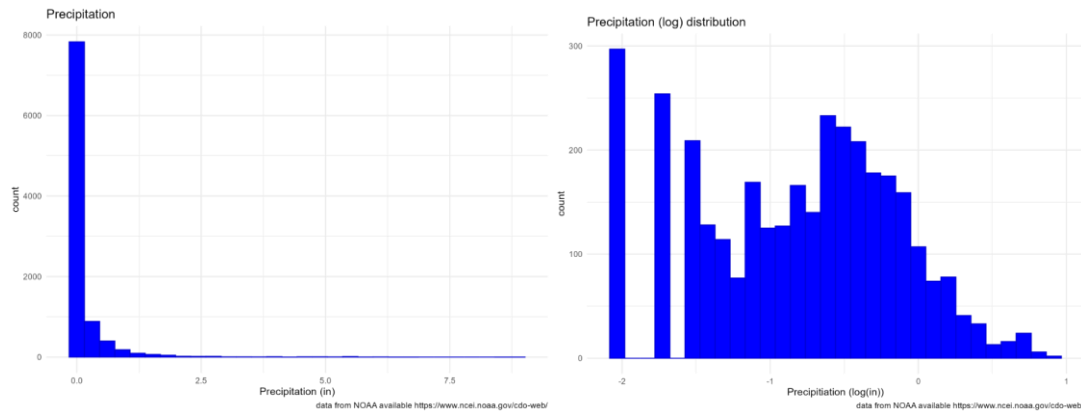
For ingestion, I loaded the datasets from my local computer. I filtered my weather dataset to include information from stations in Montgomery County. I combined the precipitation and snowfall variables into a new variable, total precipitation. I standardized the variable names to be all lowercase. And standardize the date.

Crash Data Summary:



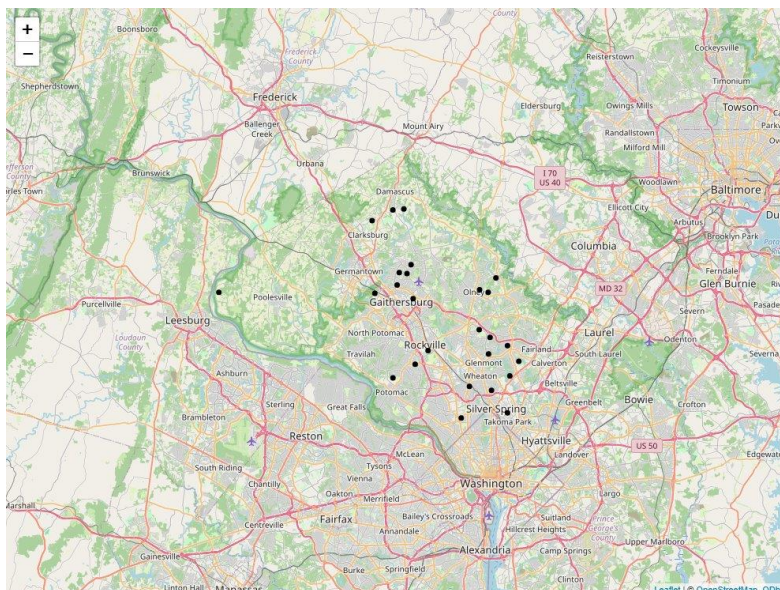
This is a map of all the crashes in 2025. The opacity is about 0.1, so any area that shows up black has a lot of crashes. Notably, the western less urban parts of the county have less crashes this is to be expected as there are fewer roads and people out there. Generally, crashes don't follow a pattern over time this can be visualized with the line graph in the EDA folder on my github. A pattern they do follow is roads. Looking at the map you can make out major road just by looking at high concentration of crashes.

Weather Data Summary:



Weather data tends to be skewed heavily to the right, which we can see in the first graph. This is due to precipitation not happening on most days of the year. Because of this, log scaling precipitation is almost always done to see variance in the data. Generally, it doesn't rain much, but there are a couple of outlier days this is due to me including snowfall into precipitation data 6 inches of snow is not unheard of where 6 inches of rain would likely cause a flash flood in the area.

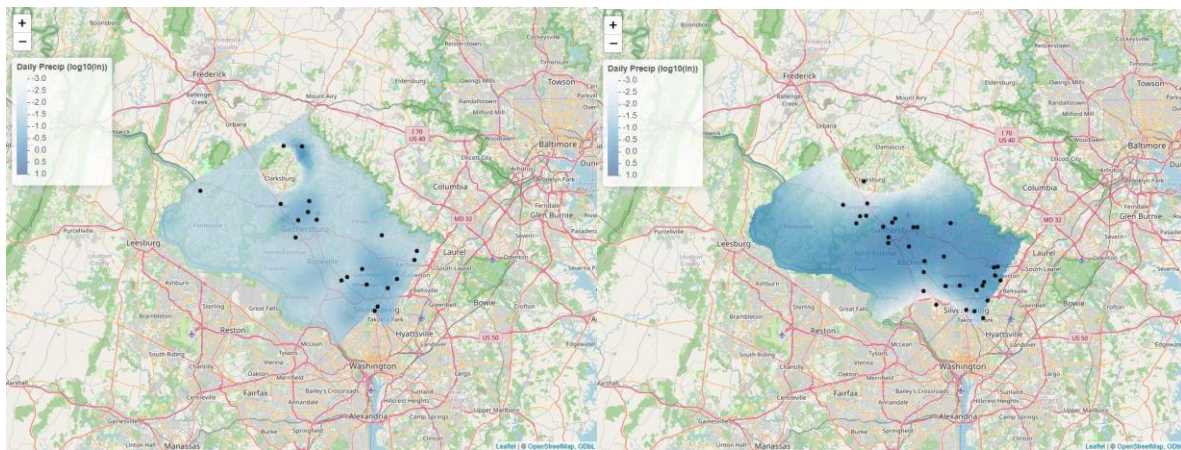
Note on Variogram and Kriging:



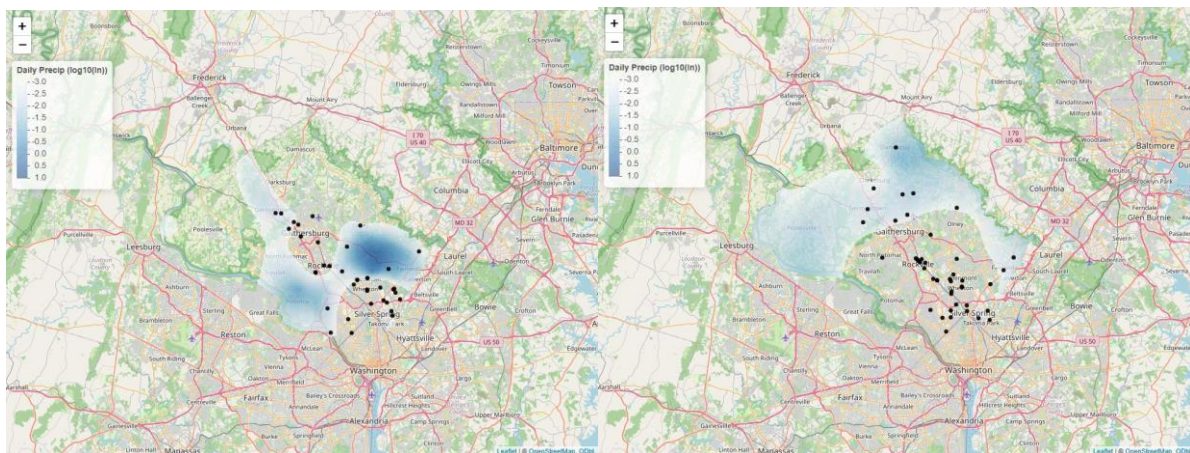
I generate the maps I show in the next section using kriging, which is a form of geospatial interpolation. The points on the map above are the weather stations in Montgomery County. Kriging is based on a Variogram model, which models spatial variability, basically how much distance and value are connected. Due to the nature of this method, it works better in areas with more data. Weather data is also very variable; it rains a lot in one place and not at all a city block over.

Map Analysis:

As mentioned previously, most days have no rain whatsoever and as such appear as just a map of crashes. Because there are still crashes when it's not raining, it means that rain is not the only factor affecting car crashes.



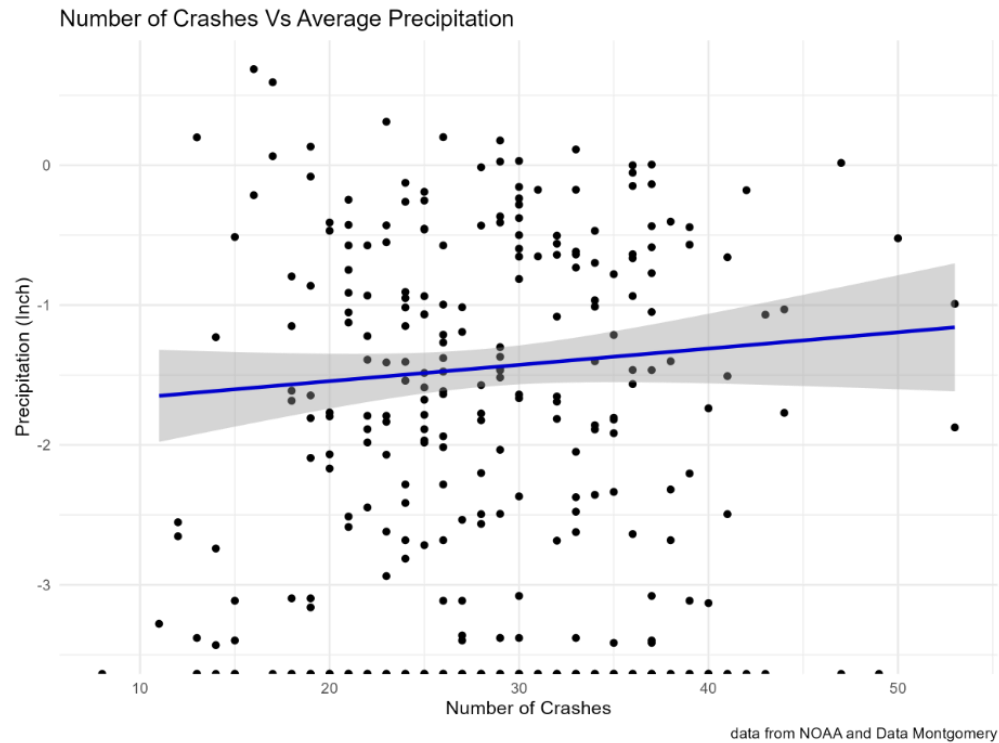
Some days, such as those above (January 1st, May 29th) do suggest a possible correlation with the areas with heavier precipitation having more crashes. With darker areas having more crashes and no crashes in the areas without rain.



Other days, such as those above (June 27th, September 19th), contradict the idea of precipitation affecting crashes. June 27th (left) does have a fairly dark patch where we would expect crashes,

but there weren't any there. September 19th had a large patch without rain in which most of the crashes on that day happened.

Correlation test:



Moving on from my maps, I conducted a correlation test with the null hypothesis that there was no correlation between average precipitation across Montgomery County and the number of crashes for a given day. I got a p-value of 0.1375, which is a fair bit above the threshold, meaning we fail to reject the null hypothesis.

Reflection and next steps:

Creating the maps using kriging was challenging, as I had never done it before. I really like how the maps came out. If I were to change anything I would expand my scope I have weather data for all of Maryland and I think it would be cool to look at. I'm on the low end for datapoints for kriging I have enough in theory, but more would likely improve the model. As a future step, I would like to find satellite weather data for a given day to compare my kriging to real data.

Acknowledgments:

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- To the early college Data Cohort

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- To friends and family outside of school for there emotional support
- And to the people who've been playing DND with me for giving me something else to fret over besides this project.